WRIGHT STATE UNIVERSITY

COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

Finding and Getting Data

Data mining and extraction · Tuesday and Thursday



Where does data come from?

Source	Examples	Good for
Public repositories	Kaggle, UCI Machine Learning Repository	Practice and learning datasets
Government portals	Data.gov, US Census Bureau	Population, economy, environment
Data libraries	Datasets built into software (scikit-learn, Orange)	Quick and clean examples
APIs	Weather, maps and other online services	Fresh data that keeps updating
Your own collection	Surveys, logs, sensors, experiments	Questions nobody else has data for

Three kinds of source

Source	How the data arrives	Example	Watch out for
Database	Records kept on purpose	Game account table	Old, duplicated or inconsistent records
Sensor	Measured automatically	Ping time, temperature	Drift, gaps, impossible readings
Survey	People answer questions	Study-habits survey	Skipped questions, untruthful answers

Ask first: which of these three did this data come from?

Structured and unstructured data

Structured

Rows and columns with named columns. Ready for Orange.

Example: a CSV of players.

Unstructured

No fixed table: text, photos, audio, video.

Example: 1,000 game reviews.

Unstructured item	One way to make it a table
A written review	Count words, or label it Positive or Negative
A photo	Measure size, brightness, or label what is in it

A good source and a weak source

Good source

Names the author or organization

Explains how the data was collected

Shows a date and a version

States the license clearly

Can be traced to the original

Weak source

Nobody is named

No description of the columns

No date, or very old

Copied from elsewhere with no link

Unclear if you may use it

Five questions to ask a source

Ask	Why it matters
Who made it?	A known organization is easier to check and to trust.
How was it collected?	A survey of 20 friends does not describe a whole country.
When was it made or updated?	Old data may describe a world that has changed.
Why was it made?	A company may only show numbers that make it look good.
Am I allowed to use it?	The license tells you what you may do with it.

Even a good source can have bad data

Missing values

Income left blank, or 999 meaning “no answer”.

Out of date

A 2010 table used to describe 2026.

Biased sample

An online poll only hears from people who answer online polls.

Mixed units

Some rows in feet, others in meters.

Duplicates and typos

The same sale listed twice. “NY” and “New York”.

Changed meaning

A column means one thing in 2015 and another in 2020.

A trusted source is not a clean dataset. Always look at the data first.

Four ways a sample goes wrong

Bias	What happens	Example
Cherry-picked data	Only rows that fit the story are kept	Showing only the best week of sales
Nonresponse	Many people asked never answer	10% reply; they differ from the rest
Voluntary response	People choose to take part	An online poll: only the angry vote
Interviewer bias	The asker changes the answer	"You enjoyed the course, right?"

A big sample does not fix a biased sample.

Which bias is it?

Situation	Which bias?
A game company reads only the reviews people chose to post.	Voluntary response
200 players got a survey and 20 answered.	Nonresponse
A chart shows only the 3 months when churn was low.	Cherry-picked data
A teacher asks "Don't you agree this class is fun?"	Interviewer bias

Name the bias first. Then say how to fix it.

How to get a dataset

Direct download

Click a link and save a file, usually CSV or Excel.

Example

Kaggle, UCI, Data.gov

Watch out

Read the description and license first. Kaggle needs a free account.

API

A website lets a program ask for data in a standard format.

Example

Today's weather for Dayton

Watch out

Often needs a key, and limits how many requests you can make.

Data library

Datasets packaged inside software.

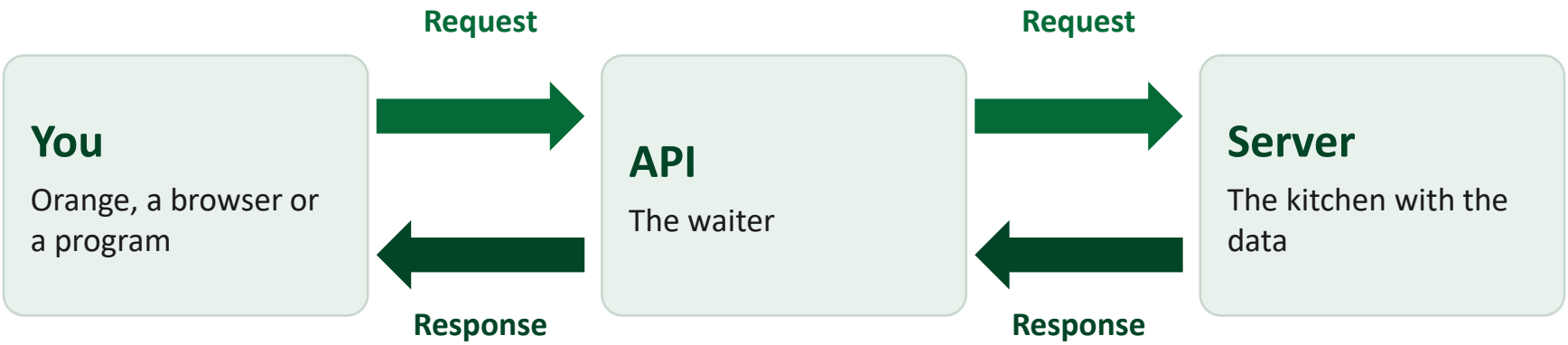
Example

scikit-learn's built-in datasets (like Iris), Orange's example data

Watch out

Small and already cleaned. Real data is messier.

How an API works



Example: “Give me today’s weather for Dayton.” → a table of temperature, rain and wind.

A checklist for a good dataset

Check	Ask yourself
Accessible format (CSV)	Can I open it in Excel or Orange right away?
Clear column names	Is the column called “Age” and not “X7”?
Consistent and low noise	Same units and spelling in every row? Few errors?
Enough rows	Are there enough rows to see a pattern?
Clear metadata	Is there a description of what each column means?
Permissive license	Does the license say I may use it this way?

Good dataset vs bad dataset

Good dataset

Student	Hours studied	Exam score
Ava	5	78
Ben	3	65
Chen	8	91
Dana	6	84

Clear names, same units, no gaps. Described, open license.

Bad dataset

c1	c2	val
NY	2019	12.5
New York	2019	12.5
ny	2020	-
LA	20/21	99999

Unclear names, mixed spellings, odd values. No description, license unknown.

With a neighbor: which checklist items does each dataset pass or fail?

Use data responsibly

License

Free to download is not free to use for anything. Read the license.

Privacy

Names, emails and locations are about real people. Do you have consent?

Terms of service

Some websites do not allow you to copy their data.

Fairness

Who is missing from the data? Who might be harmed by it?

Data augmentation

What it is

Making new, slightly changed examples from the data we already have.

Synthetic data is made by a program, not collected from the world.

Used for: class imbalance

One group has far fewer examples, like fraud or a rare defect.

Used for: overfitting

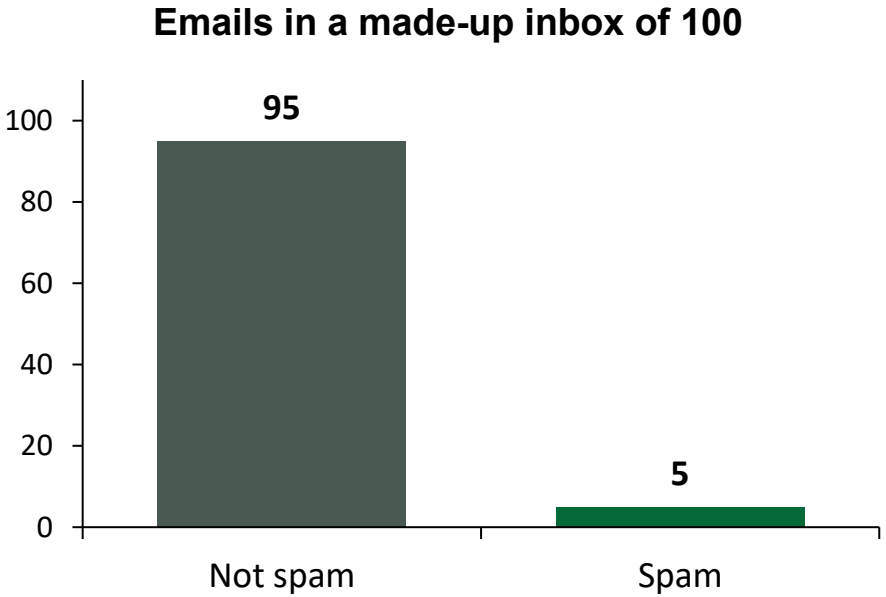
A model memorizes a small dataset instead of learning the pattern.

Class imbalance: one group has far fewer examples than another.
Overfitting: a model memorizes its practice data, then fails on new data.

Why we augment data

Fix class imbalance	Give the rare group enough examples to learn from.
Reduce overfitting	More variety, so the model learns the pattern.
Handle messy real life	Photos are tilted, and people word things differently.
Small datasets	Only a few hundred rows? Augmentation adds variety.
Save time and money	Collecting more real data can be slow or costly.

Why class imbalance is a problem



Always guess "not spam"?

95% accurate.

0 spam emails found.

Accuracy hides the problem.

Fix: give the model more spam examples to learn from.

Class imbalance in real life

Situation	Common class	Rare class (we care)	Rough split
Card payments	Normal purchase	Fraud	999 : 1
Factory parts	Good part	Defective part	98 : 2
Email	Not spam	Spam in the inbox	95 : 5
Machine sensors	Running fine	About to fail	99 : 1
Course records	Passed	Dropped out	90 : 10

Why imbalance fools a model

The model mostly sees the common class.

Always guessing "common" already scores high.

The rare cases are missed, and they matter most.

1,000 payments	Model said normal	Model said fraud
990 really normal	990	0
10 really fraud	10	0

"Always normal" model

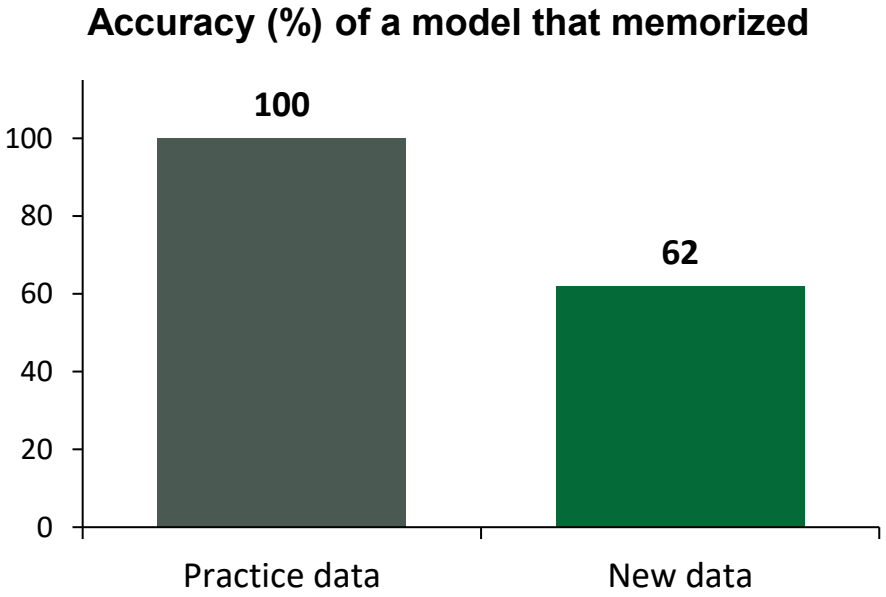
Accuracy: 99%

Fraud caught: 0 of 10

Four ways to handle imbalance

Collect more	Get more real rare examples. Best, but often hard.
Augment the rare class	Make new rare examples (SMOTE, ADASYN).
Remove common rows	Use fewer common rows. Loses real data.
Better measures	Count the rare cases caught, not just accuracy.

What overfitting means



Overfitting

Great on practice data.

Poor on new data.

The model memorized instead of learned.

Fix: add variety so memorizing stops working.

Overfitting in real life

Situation	What it memorized	What goes wrong
Exam study	Answers to one practice test	Fails a new test
Face unlock	20 photos in one room	Fails in new lighting
Spam filter	Exact words of old emails	Misses reworded spam
Grade predictor	Student ID numbers	No pattern for new students
Defect detector	Parts from one machine	Fails on other machines

Why small datasets overfit

The model sees only a few examples.

It memorizes their quirks and noise.

New data looks different, so it fails.

Model	Practice data	New data
Memorizer	100%	62%
Pattern learner	91%	89%

Look at the gap

Memorizer: 38 points lower on new data

Pattern learner: 2 points lower

Four ways to reduce overfitting

Collect more	Get more real examples. Best, but often costly.
Augment the training data	Add varied copies so memorizing stops working.
Use fewer columns	Drop IDs and repeated columns (Weeks 5 and 6).
Check on new data	Compare practice and new scores (Week 7).

Techniques by data type

Data type	Technique	Plain idea	Helps with
Tables	SMOTE	Synthetic Minority Over-sampling Technique: creates new rows between two real rare rows.	Class imbalance
Tables	ADASYN	Adaptive Synthetic sampling: like SMOTE, with more new rows where learning is hardest.	Class imbalance
Tables	Noise injection	Adds a tiny random wiggle to numbers (70.0 becomes 70.3).	Overfitting
Text	Back-translation	Translate to another language and back to get new wording.	Overfitting
Text	Synonym replacement	Swap words for similar ones (“big” becomes “large”).	Overfitting
Images	Colors, angles, cropping	Flip, rotate, crop or recolor the same picture.	Overfitting

SMOTE: make new rare rows

What it does

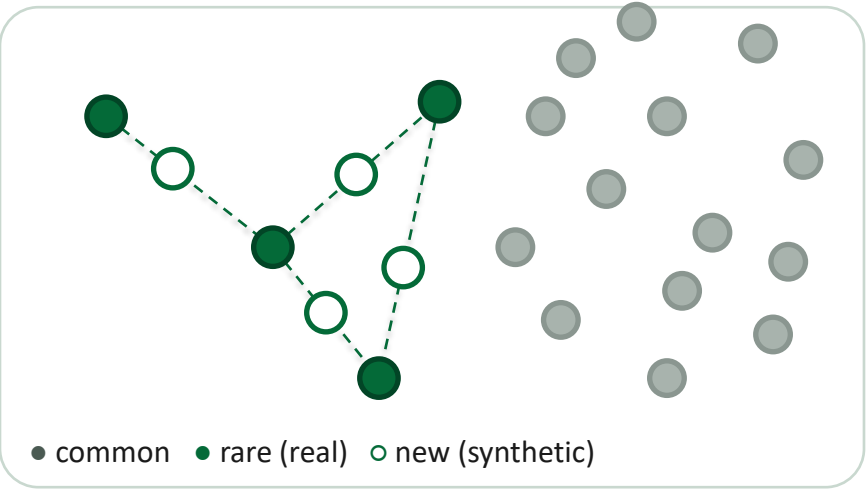
Creates new rare rows between two real rare rows.

Why use it

Balances the classes without copying rows.

Watch out

The new row is a blend, not a real case.



Example: rare A = (2, 4), rare B = (4, 8). Halfway = (3, 6) = new row.

ADASYN: more help where it is hard

What it does

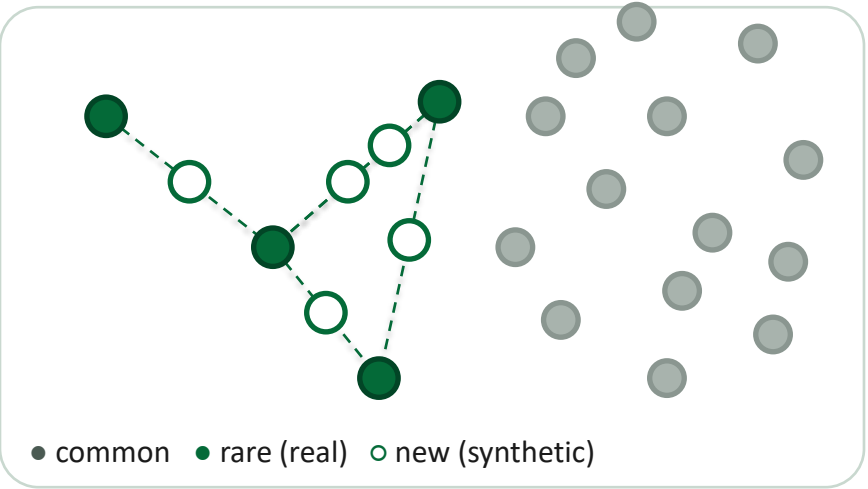
Like SMOTE, but adds more rows where the rare class is hard.

Why use it

Helps most at the border between the two groups.

Watch out

Can also copy noise or mistakes near the border.



Rare points close to common ones get more new points.

Noise injection: small wiggles

What it does

Adds a tiny random change to each number.

Why use it

Makes copies that are similar, not identical.

Watch out

Too much noise makes impossible rows.

Row	Temp (°C)	Price ($)
Original	70.0	4.99
New copy 1	70.3	5.02
New copy 2	69.8	4.97

Same meaning, tiny difference. Works for numeric columns only.

Back-translation: new wording

What it does

Translates a sentence to another language and back.

Why use it

Gives new wording with the same meaning.

Watch out

Translations can slip. Read some results.

English (original)

The movie was great.

translate

Spanish (middle step)

La película fue genial.

translate back

English (new version)

The film was really good.

Synonym replacement: swap words

What it does

Swaps a few words for similar words.

Why use it

Quick and cheap way to get new sentences.

Watch out

Words with two meanings can change the sense.

Original (positive review)

The food was good and the service was fast.

New version (still positive)

The **meal** was **great** and the service was **quick**.

Changed words are shown in green.

Images: rotate and flip

What it does

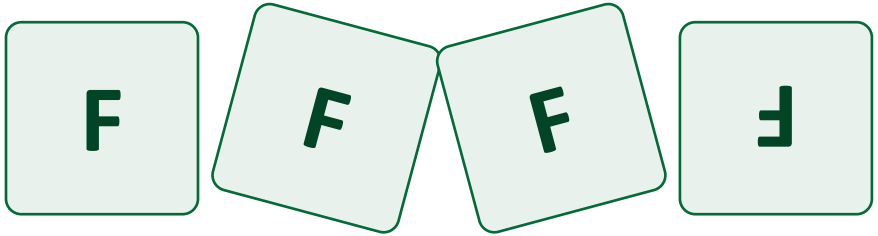
Tilts or flips the picture by a small angle.

Why use it

The model learns objects seen from many angles.

Watch out

Some turns change meaning, like 6 and 9.



Warning: a 6 upside down looks like a 9, so the label would be wrong.

Images: cropping

What it does

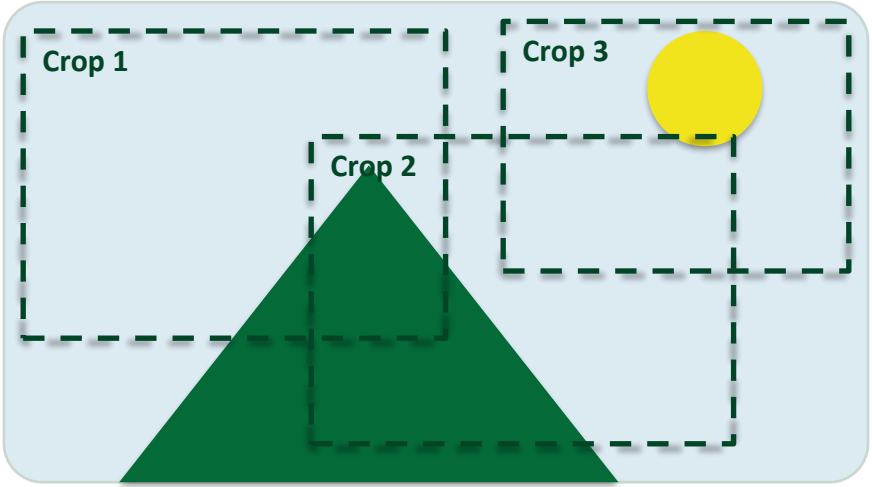
Cuts out a different part of the same picture.

Why use it

The object can sit anywhere in the frame.

Watch out

Do not crop out the thing you are labeling.



Each crop is a new picture of the same scene.

Images: change colors

What it does

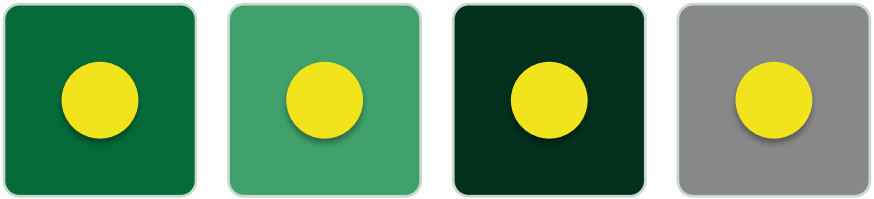
Changes brightness, tint or turns the picture gray.

Why use it

Photos are taken in different light and cameras.

Watch out

Do not do it when color is the clue (ripe fruit).



Original Brighter Darker Gray

Same subject, different lighting. The label does not change.

Augmentation: do and don't

Do

Augment the training data only, never the test data.

Keep labels true: a rotated 6 can look like a 9.

Keep the original data too.

Check that new examples look realistic.

Don't

Treat synthetic rows as new facts. They add variety, not information.

Use it to rescue a tiny, poor dataset.

Create impossible rows, like a 300-year-old customer.